A BRIEF INTRODUCTION INTO MODEL CATEGORIES

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1. Introduction

The theory of model categories, introduced by Daniel Quillen in his 1967 treatise [Qui67], formalizes the ideas of homotopy theory from topology and homological algebra in a general category-theoretic setting. Quillen's introduction of model categories both elucidated homotopy theory in the classical setting of algebraic topology as well as brought homotopical algebra to a wide-range of areas at the intersection of algebra and geometry (see for example [Dug22], [MV99], [HHR16]).

The following is a brief exposition on the basic definitions and constructions one first encounters when learning model categories. All of the material is taken from [DS95].

2. Model categories

We begin by fixing some notation.

Notation 2.1.

- We denote by R some fixed commutative unital ring, and we write $\mathsf{Ch}_{\geq 0}(R)$ for the category of chain complexes of (right) R-modules concentrated in non-negative degrees.
- We let Top denote the category of *compactly generated*[1] spaces and continuous maps between them.

Before formally defining the notion of a "model category structure", a few remarks must be made on some category-theoretic terminology.

Remark 2.2 (A word on (co)limits)**.** The definition of a model category M requires M to have all *small*[2] limits and colimits. For those unfamiliar with the notion of *(co)limits* in category theory, the fastest way to develop a working intuition of these concepts is via some notable examples:

- products[3] and inverse limits are examples of limits, as well as kernels in additive categories.
- coproducts and direct limits are examples of colimits, as well as cokernels in additive categories.

A readable introduction to the theory of limits and colimits for the purpose of learning model categories is given in section 2 of [DS95].

We will take for granted the following result concerning the existence of limits and colimits in the categories central to classical homotopy theory.

Fact 2.3. The categories $\mathsf{Ch}_{\geq 0}(R)$ and Top have all small limits and colimits.

The definition of a model category structure also requires the notion of *retracts*.

Definition 2.4. Given morphisms $f : A \to B$ and $g : C \to D$ in a category C, we say f is a **retract** of g if there exists a commutative diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d, "f"'] \arrow[rr, bend left, equal] & C \arrow[r] \arrow[d, "g"] & A \arrow[d, "f"] \\ B \arrow[r] \arrow[rr, bend right, equal] & D \arrow[r] & B \end{tikzcd}$$

in C.

Remark 2.5. We could have defined a *retract* of an object A in C to be an object B such that 1_B admits a factorization through A (i.e. the usual notion of a retract from topology). Then f would be a retract of g in this classical sense in the *arrow category* $\mathsf{Arr}(\mathsf{C})$ of C, whose objects are the morphisms of C and with morphisms given by commutative squares in C.

The following terminology captures properties which are ubiquitous throughout the theory of model categories.

[1] It is not important for the reader to know what this means. A good reference to compactly generated spaces is [Rez18]. The key takeaways are that this category contains all of the "nice spaces" one is interested in and, unlike the category of *all* topological spaces, it is cartesian closed, i.e. $- \times X$ has a right adjoint for every space X.

[2] A category C is said to be small if its collection of objects $\mathsf{ob}(\mathsf{C})$ form a proper set. Formally limits and colimits are functors, and we say they are small if their source is a small category.

[3] This includes terminal objects, as they are the same as an empty product, and similarly initial objects are empty coproducts.

Definition 2.6. Given morphisms $i : X \to Y$ and $p : A \to B$ in C, we say that i has the **left lifting property** with respect to p, or equivalently that p has the **right lifting property** with respect to i, denoted $i \boxslash p$, if for every commutative square

$$\begin{tikzcd} X \arrow[r] \arrow[d, "i"'] & A \arrow[d, "p"] \\ Y \arrow[r] \arrow[ur, dashed, "h"] & B \end{tikzcd}$$

in C, there exists a dashed lift $h : Y \to A$ making the above diagram commute. Given two collections of morphisms $\mathcal{L}, \mathcal{R}$ in C, we say $\mathcal{L}$ has the **left lifting property** with respect to $\mathcal{R}$, or equivalently $\mathcal{R}$ has the **right lifting property** with respect to $\mathcal{L}$, if $f \boxslash g$ for every $f \in \mathcal{L}$ and every $g \in \mathcal{R}$.

We are now ready to define model categories.

Definition 2.7. A **model structure** on a category M consists of three classes of distinguished morphisms in M: **weak equivalences** $\mathcal{W}$, **cofibrations** $\mathcal{C}$, and **fibrations** $\mathcal{F}$. This data is subject to the following axioms:

- M1 (2-out-of-3): Given a pair of composable arrows $A \xrightarrow{f} B \xrightarrow{g} C$ in M, if two of f, g, gf are weak equivalences, then so is the third.
- M2 (Retracts). Weak equivalences, cofibrations, and fibrations are closed under retracts.
- M3 (Lifting properties). The cofibrations have the left lifting property with respect to the **acyclic fibrations** $\mathsf{W} \cap \mathcal{F}$, and the fibrations have the right lifting property with respect to the **acyclic cofibrations** $\mathcal{W} \cap \mathcal{C}$.
- M4 (Factorization). Any morphism $f : A \to B$ in M can be factored as

$$f : A \xrightarrow{i} \widetilde{A} \xrightarrow{p} B$$

where i is an acylic cofibration and p is a fibration, and

$$f : A \xrightarrow{i} \widetilde{B} \xrightarrow{p} B$$

where i is a cofibration and p is an acylic fibration.

Definition 2.8. A **model category** is a category M which has a model structure and contains all small limits and colimits.

Convention 2.9. Throughout, M will denote an arbitrary fixed model category.

Notation 2.10. We denote weak equivalences by $\xrightarrow{\sim}$, cofibrations by $\rightarrowtail$, and fibrations by $\twoheadrightarrow$.

The existence of an initial object $\varnothing$ and terminal object $*$ in M lead to the following important definitions.

Definition 2.11. We say an object A in M is **fibrant** if the unique morphism $A \to *$ is a fibration. Dually, we say A is **cofibrant** if $\varnothing \to A$ is a cofibration. If A is both cofibrant and fibrant, we say A is **cofibrant-fibrant**. We denote the subcategories of M spanned by fibrant, cofibrant, and cofibrant-fibrant objects by M^f, M^c, and M^{cf} respectively.

While in general not every object in M will be co/fibrant, the factorization axiom applied to $A \to *$ (resp. $\varnothing \to A$) guarantees that every object A is weakly equivalent to a fibrant (resp. cofibrant) object.

Definition 2.12. Given an object A in M, an acyclic fibration $QA \xrightarrow{\sim}\!\!\!\!\twoheadrightarrow A$ with QA cofibrant is called a **cofibrant replacement** of A. A **fibrant replacement** of A is an acyclic cofibration $A \rightarrowtail\!\!\!\!\xrightarrow{\sim} RA$ with RA fibrant.

The axioms of a model category structure guarantee not only the existence of cofibrant and fibrant replacements, but cofibrant-fibrant replacements as well.

Proposition 2.13. *Given an object A in M, RQA is cofibrant, hence in particular it is cofibrant-fibrant. A similar statement holds for QRA.*

Proof. This proof relies on a standard result, Proposition 3.13 [DS95], which characterizes cofibrations precisely as those morphisms which have the left lifting property with respect to acylic fibrations.

We want to show the map $\varnothing \to RQX$ is a cofibration, hence we want to solve the following lifting problem:

$$\begin{tikzcd} \varnothing \arrow[r] \arrow[d] & X \arrow[d, two heads, "p", "\sim"'] \\ RQA \arrow[r, "f"'] & Y. \end{tikzcd}$$

Since RQA is a fibrant replacement of QA, which itself is cofibrant, we have a factorization $\varnothing \rightarrowtail QA \rightarrowtail\!\!\!\!\xrightarrow{\sim} RQA$. Denoting the acylic cofibration in the composition by q, the lifting axiom guarantees the existence of a morphism $\widetilde{h} : QA \to X$ such that diagram

$$\begin{tikzcd} \varnothing \arrow[r] \arrow[d, tail] & X \arrow[d, two heads, "p", "\sim"'] \\ QA \arrow[r, "f \circ q"'] \arrow[ur, dashed, "\widetilde{h}"] & Y \end{tikzcd}.$$

Applying the lifting axiom again to the commutative square

$$\begin{tikzcd} QA \arrow[r, "\widetilde{h}"] \arrow[d, tail, "q", "\sim"'] & X \arrow[d, two heads, "p", "\sim"'] \\ RQA \arrow[r, "f"'] & Y \end{tikzcd}$$

gives us our desired lift $h : RQA \to X$.

□

2.1. **Examples of Model Categories.**

Example 2.14 (The opposite category)**.** Indeed, like the structure of a preadditive category, the axioms of a model category structure are self-dual. That is, if $\mathcal{W}, \mathcal{C}, \mathcal{F}$ denote the weak equivalences, cofibrations, and fibrations respectively of M, then M^{op} is a model category with weak equivalences given by $\mathcal{W}^{\mathrm{op}}$, cofibrations given by $\mathcal{F}^{\mathrm{op}}$, and fibrations given by $\mathcal{C}^{\mathrm{op}}$. This is the precise way in which any statement made involving the cofibrations of M has an associated dual statement in terms of fibrations, viewed as cofibrations of M^{op}.

Example 2.15 (Spaces)**.** The most common model category structure which generalizes the classical homotopy theory of topological spaces is the **Quillen model structure** on the category Top:

- Weak equivalences are the weak homotopy equivalences, i.e., continuous maps which induce isomorphisms on homotopy groups.
- Cofibrations are maps $f : X \to Y$ such that f is a retract of a map $X \to Y'$ with Y' a relative CW complex of X.
- Fibrations are Serre fibrations, i.e. those maps which have the right lifting property with respect to inclusions of disks $i_0 : D^n \to D^n \times I$, $x \mapsto (x, 0)$.

Every object is fibrant with respect to this model structure. Cofibrant objects are those spaces which are retracts of *generalized CW complexes* (i.e. spaces X which are built up from cells but do not have to be constructed sequentially via skeleta X_n indexed by the dimension $n \in \mathbf{N}$.)

Example 2.16 (Chain complexes)**.** There is a model category structure on $\mathsf{Ch}_{\geq 0}(R)$, called the **projective model structure**, where

- weak equivalences are quasi-isomorphisms;
- cofibrations are degree-wise monomorphisms whose cokernel is a chain complex of projectives.
- fibrations are degree-wise epimorphisms

It is straightforward to understand what cofibrant and fibrant chain complexes are in this model structure. Given $C_\bullet \in \mathsf{Ch}_{\geq 0}(R)$, $C_\bullet \to 0$ is always a degree-wise epimorphism, hence every object is fibrant. Dually, as $C_\bullet \cong \operatorname{coker}(0 \to C_\bullet)$, the cofibrant objects are precisely the chain complexes of projective R-modules. Given an R-module M, denoting by $M[0]$ the chain complex concentrated in degree 0 with $M[0]_0 = M$, note a projective resolution of M is equivalent to the data of a chain map

$$\begin{tikzcd} \dots \arrow[r] & P_2 \arrow[r, "d_2"] \arrow[d] & P_1 \arrow[r, "d_1"] \arrow[d] & P_0 \arrow[r] \arrow[d, "\varepsilon"] & 0 \arrow[r] \arrow[d] & \dots \\ \dots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & M \arrow[r] & 0 \arrow[r] & \dots, \end{tikzcd}$$

with $P_\bullet$ an acylic complex of projectives. This implies in particular that such a chain map is a quasi-isomorphism, hence projective resolutions of R-modules are precisely cofibrant replacements with respect to the projective model structure.

Remark 2.17. One may notice that in both concrete examples of model category structures given above, the fibrations were comparatively simpler to describe than the cofibrations. This stems from the fact that both of these model structures are *cofibrantly generated*, i.e., the collections of weak equivalences, cofibrations, and fibrations are all "generated" in some sense from two initial *sets* I and J of cofibrations and acyclic cofibrations. Somewhat counterintuitively, the fibrations of a cofibrantly generated model category are constructed from I and J in fewer "steps" than the cofibrations, leading to the discrepancy in their definitions.

3. The homotopy category of a model category

3.1. **The homotopy category via the homotopy relation on maps.**

If model categories are to serve as an appropriate framework by which to generalize homotopy theory, they should come with an appropriate notion of a homotopy category. In the classical homotopy theory of topological spaces, we construct $\mathsf{Ho}(\mathsf{Top})$ as a category consisting of the same objects as those in Top and with morphism sets $\mathrm{Hom}_{\mathsf{Ho}(\mathsf{Top})}(X, Y)$ defined as the quotient $\mathsf{Top}(X, Y)/\sim$, where $f \sim g$ is the usual (left) homotopy relation between continuous maps. The key observation is that a homotopy $H : X \times I \to Y$ from f to g is

equivalent to the existence of a commutative diagram

$$\begin{tikzcd} A \coprod A \arrow[r, "i_0 \sqcup i_1"] \arrow[dr, "f \sqcup g"'] & A \times I \arrow[d, dashed, "H"] \\ & B \end{tikzcd}$$

where $i_k : a \mapsto (a, k)$. The cartesian-closed structure of Top, i.e. the existence of an adjunction $\mathsf{Top}(A \times B, C) \cong \mathsf{Top}(A, C^B)$[4], allows us to equivalently define a homotopy H via the existence of a commutative diagram

$$\begin{tikzcd} B \times B & B^I \arrow[l, "p_0 \times p_1"'] \\ & A, \arrow[ul, "f \times g"] \arrow[u, dashed, "H"'] \end{tikzcd}$$

where $p_0 : \gamma \mapsto \gamma(0)$ and $p_1 : \gamma \mapsto \gamma(1)$ take a path to its initial point and terminal point respectively. The existence of products and coproducts gives us a nice generalization of these constructions to model categories.

Definition 3.1. Let A be an object in M.

1. A **cylinder object** for A consists of the data of a factorization of the codiagonal morphism $\nabla : A \coprod A \to A$ via a cofibration followed by a trivial fibration:

$$\begin{tikzcd} A \coprod A \arrow[rr, "\nabla"] \arrow[dr, tail, "i_0 \sqcup i_1"'] & & A \\ & \mathrm{cyl}(A) \arrow[ur, two heads, "\sim"'] & \end{tikzcd}.$$

2. A **path object** for A consists of the data of a factorization of the diagonal morphism $\Delta : A \to A \times A$ as a trivial cofibration followed by a fibration:

$$\begin{tikzcd} A \arrow[rr, "\Delta"] \arrow[dr, tail, "\sim"'] & & A \times A \\ & \mathrm{path}(A) \arrow[ur, two heads, "p_0 \times p_1"'] & \end{tikzcd}.$$

3. Two morphisms $f, g : A \to B$ in M are **left homotopic**, denoted $f \sim_l g$, if there exists a morphism $H : \mathrm{cyl}(A) \to B$ such that $f \sqcup g = H \circ (i_0 \sqcup i_1)$.
4. We say $f, g : A \to B$ are **right homotopic**, denoted $f \sim_r g$, if there exists a morphism $H : A \to \mathrm{path}(B)$ such that $f \times g = (p_0 \times p_1) \circ H$.

Proposition 3.2. *Let A, B be objects in M*

1. *If A is cofibrant then $\sim_l$ is an equivalence relation on $\mathrm{Hom}_{\mathsf{M}}(A, B)$.*
2. *If B is fibrant then $\sim_r$ is an equivalence relation on $\mathrm{Hom}_{\mathsf{M}}(A, B)$.*
3. *If both A is cofibrant and B is fibrant then left homotopy $\sim_l$ and right homotopy $\sim_r$ coincide, and we denote this unique equivalence relation on $\mathrm{Hom}_{\mathsf{M}}(A, B)$ by $\sim$.*

In particular, on the subcategory M^{cf} of M consisting of cofibrant-fibrant objects, we have a singular notion of homotopy between morphisms. This gives us the key idea to define the homotopy category of a model category.

Definition 3.3. The **homotopy category** $\mathsf{Ho}(\mathsf{M})$ of a model category M is defined as the category with the same objects as M and, given $A, B \in \mathsf{M}$,

$$\mathrm{Hom}_{\mathsf{Ho}(\mathsf{M})}(A, B) \coloneqq \mathrm{Hom}_{\mathsf{M}^{cf}}(RQA, RQB)/\sim .$$

3.2. **The homotopy category via localization.** There is an equivalent formulation of the classical homotopy categories of Top and $\mathsf{Ch}_{\geq 0}(R)$ which is less concrete but provides a stronger conceptual justification for why model categories provide the correct setting to "do" homotopy theory. This approach is centered on the idea of the *localization* of a category M, an analogous construction to the localization of ring, where instead of inverting a multiplicative subset, we "formally invert" a class of morphisms in M.

Definition 3.4. Given a category C and collection of morphisms $\mathcal{W}$ in C, the **localization** of C with respect to $\mathcal{W}$ consists of a category $\mathsf{C}[\mathcal{W}^{-1}]$ along with a functor $F : \mathsf{C} \to \mathsf{C}[\mathcal{W}^{-1}]$ satisfying the following properties:

1. For every morphism $f \in \mathcal{W}$, $F(f)$ is an isomorphism in $\mathsf{C}[\mathcal{W}^{-1}]$.

[4]Here C^B denotes an internal function object of Top, i.e. the set $\mathsf{Top}(B, C)$ topologized in an appropriate way.

2. F satisfies the following universal property: given a functor $G : \mathsf{C} \to \mathsf{D}$ such that for every morphism $f \in \mathcal{W}$, $G(f)$ is an isomorphism, G factors uniquely along F:

$$\begin{tikzcd} \mathsf{C} \arrow[r, "G"] \arrow[d, "F"'] & \mathsf{D} \\ \mathsf{C}[\mathcal{W}^{-1}] \arrow[ur, dashed, "\exists!"'] & \end{tikzcd}.$$

Warning 3.5. While the definition of the localization of a category seems quite natural, in general one should not expect localizations to actually exist. This is because, proceeding with our analogy with localizations of rings, given a category C and a class of morphisms $\mathcal{W} \in \mathrm{mor}(\mathsf{C})$, we can model $\mathsf{C}[\mathcal{W}^{-1}]$ as a category with the same objects as C, and whose collection of morphisms is given by an enlargement of $\mathrm{mor}(\mathsf{C})$ in which we adjoin formal inverses to elements of $\mathcal{W}$ and take the "closure" with respect to compositions.[5] In general, even if $\mathcal{W}$ is a proper set, there is no reason the "hom-sets" of $\mathsf{C}[\mathcal{W}^{-1}]$ will actually be sets, hence the need to enlarge our set-theoretic universe.

There are some important instances where localizations of categories do in fact exist.

Example 3.6. The homotopy category $\mathsf{Ho}(\mathsf{Top})$ is isomorphic to the localization of Top with respect to the class of homotopy equivalences. In practice, most algebraic topologists work in the homotopy category $\mathsf{Ho}(\mathsf{CW})$, where CW denotes the full subcategory spanned by CW complexes. Cellular approximation[6] together with Whitehead's theorem[7] give an equivalence of categories between $\mathsf{Ho}(\mathsf{CW})$ and the localization of Top with respect to the class of weak homotopy equivalences.

This example illustrates an important feature of general model categories: they provide a sufficient structure to guarantee the existence of localizations:

Theorem 3.7. *The homotopy category* $\mathsf{Ho}(\mathsf{M})$ *defined above is the localization* $\mathsf{M}[\mathcal{W}^{-1}]$ *of* M *with respect to the collection* $\mathcal{W}$ *of weak equivalences.*

Example 3.8. The above theorem allows us to conclude that the homotopy category of $\mathsf{Ch}_{\geq 0}(R)$ with the projective model structure is in fact the derived category $\mathsf{D}(R)$ of R. An analogous statement is true when Mod_R is replaced by a general abelian category.

4. A Brief Word on Derived Functors

Model categories provide a useful setting to develop the theory of derived functors. In the context of a functor F defined on a model category M, the left or right derived functor of F is simply a best approximation (in some universal sense) to an "extension" of F along the localization functor $\gamma : \mathsf{M} \to \mathsf{Ho}(\mathsf{M})$.[8] We present the theory of left derived functors. The theory of right derived functors follows by duality.

Definition 4.1. Let D be a (not necessarily model) category and $F : \mathsf{M} \to \mathsf{D}$ be a functor. A **left derived functor** LF for F consists of a functor $LF : \mathsf{Ho}(\mathsf{M}) \to \mathsf{D}$ and a natural transformation $\eta : LF \circ \gamma \Rightarrow F$ satisfying the following universal property: given another pair (G, ε) of a functor $G : \mathsf{Ho}(M) \to \mathsf{D}$ and a natural transformation $\varepsilon : G \circ \gamma \Rightarrow F$, there exists a unique natural transformation $\widetilde{\varepsilon} : G \Rightarrow LF$ such that $\varepsilon = \eta \circ \widetilde{\varepsilon}\gamma$, where $\varepsilon\gamma : G \circ \gamma \Rightarrow LF \circ \gamma$ is the natural transformation defined by $(\varepsilon\gamma)_A \coloneqq \varepsilon_{\gamma(A)}$ for $A \in \mathsf{M}$.

IF D is also a model category, there is a class of derived functors of particular interest.

Definition 4.2. Given a functor $F : \mathsf{M} \to \mathsf{N}$ between model categories, a **total left derived functor** $\mathbf{L}F$ of F is a left derived functor of $\gamma_D \circ F$.

Example 4.3. The classical left and right derived functors of Tor and Ext from homological algebra are examples of total derived functors in the above sense. More details on this can be found in Chapter 9 of [DS95].

References

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[5] If we wanted to make this loose analogy more rigorous, we would also need to "mod out" in some appropriate sense the associativity and identity relations which are required in defining a category.

[6] Every topological space admits a weak homotopy equivalence to a CW complex.

[7] Weak homotopy equivalences between CW complexes coincide with homotopy equivalences.

[8] For those familiar with Kan extensions, we have $LF \coloneqq \mathsf{Ran}_\gamma F$ and $RF \coloneqq \mathsf{Lan}_\gamma F$.